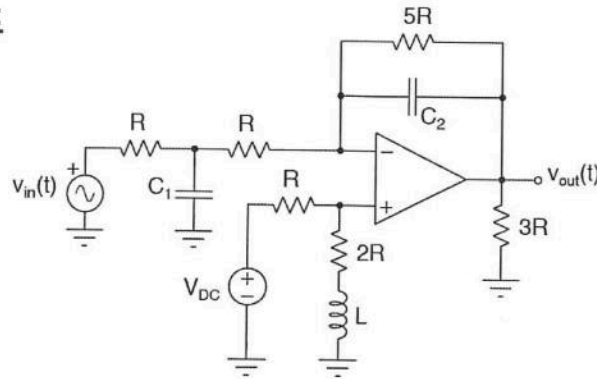


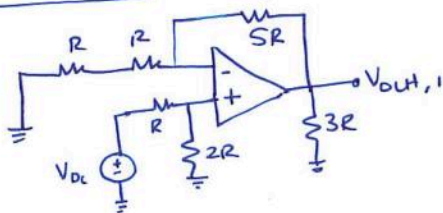
Question 1 [25 Points]:



In the circuit above, $R = 1 \text{ k}\Omega$, $L = 1 \text{ mH}$, $C_1 = 3 \text{ nF}$, $C_2 = 0.4 \text{ nF}$. Assume that the OPAMP is ideal and not saturated, $v_{in}(t) = 2 \cos(10^6 t) \text{ V}$ and $V_{DC}(t) = 1 \text{ V (DC)}$, for $-\infty < t < \infty$. Find $v_{out}(t)$.

Use superposition:

* From V_{DC} : capacitor $\Rightarrow$ open circuit, inductor $\Rightarrow$ short circuit

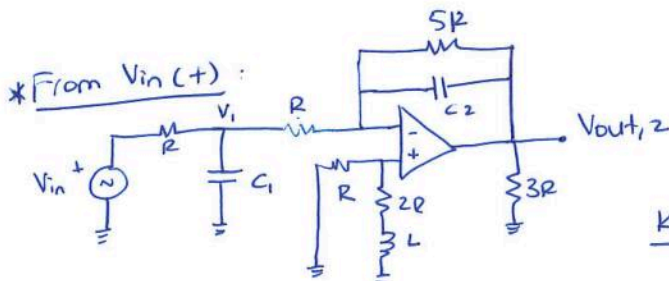


$$V_+ = V_{DC} \cdot \frac{2R}{3R} = \frac{2}{3} \text{ V (voltage divider)}$$

$$V_+ = V_- = \frac{2}{3} \text{ V}$$

$$V_- = V_{out,1} \cdot \frac{2R}{2R + 5R} \quad (\text{voltage divider})$$

$$V_{out,1} = \frac{7}{2} V_- = \frac{7}{2} \cdot \frac{2}{3} = \boxed{\frac{7}{3} \text{ V (DC)}}$$



$$V_{in} = 2 \quad (\text{in phasor form})$$

$$V_+ = V_- = 0 \text{ V (no input on } V_+ \text{ side)}$$

$$\text{KCL @ } V_1: \frac{V_1}{\frac{1}{j\omega C_1}} + \frac{V_1 - V_{in}}{R} + \frac{V_1}{R} = 0$$

$$V_1(j\omega C_1 R + 2) = V_{in}$$

$$V_1 = \frac{V_{in}}{2 + j\omega C_1 R}$$

$$\text{KCL @ } V_-: -\frac{V_1}{R} - \frac{V_{out,2}}{5R} - \frac{V_{out,2}}{\frac{1}{j\omega C_2}} = 0$$

$$V_{out,2}(j\omega C_2 \cdot 5R + 1) = -5V_1 = -\frac{5V_{in}}{2 + j\omega C_1 R}$$

$$V_{out,2} = -\frac{5V_{in}}{(1 + j5\omega R C_2)(2 + j\omega R C_1)} = -\frac{5 \cdot 2}{(1 + j5 \cdot 10^6 \cdot 10^{-3} \times 0.4 \times 10^{-9})(2 + j10^6 \cdot 10^{-3} \cdot 3 \cdot 10^{-9})}$$

$$= -\frac{10}{(1 + j2)(2 + j3)} = \frac{-10}{-4 + j7} = \frac{10}{4 - j7} = \frac{10}{\sqrt{65} \angle -\tan^{-1}(7/4)}$$

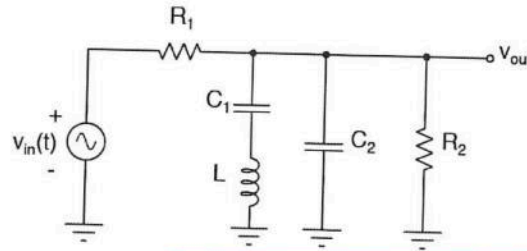
$$= \frac{10}{\sqrt{65}} e^{j \tan^{-1}(7/4)}$$

$$V_{out,2}(t) = \frac{10}{\sqrt{65}} \cos(10^6 t + \tan^{-1}(7/4)) \text{ V} = \boxed{1.24 \cos(10^6 t + 1.05) \text{ V}}$$

$\phi = 1.05 \text{ rad, or } 60.3^\circ$

$$\text{* Finally, } V_{out}(t) = V_{out,1} + V_{out,2}(t) = \boxed{\frac{7}{3} + 1.24 \cos(10^6 t + \phi) \text{ V}}$$

Question 2 [25 Points]:



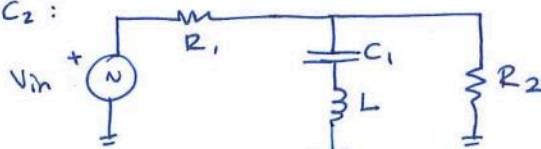
- (a) [6 points] In the circuit shown above, first ignore C_2 (i.e., replace it with open circuit). What is the resonance frequency ω_0 of the circuit? What is the Q of the circuit? Express ω_0 and Q in parametric form first (without inserting component values). **Hint:** You may want to use source transformation (i.e., Thevenin and/or Norton equivalent circuits) to find Q.
- (b) [11 points] This circuit actually has two resonance frequencies, ω_1 and ω_2 . Find ω_1 and ω_2 in parametric form (without inserting component values). **Do not** ignore C_2 in these derivations. **Hint:** $\text{Im}\{Z_{eq}(\omega)\} = 0$ at resonance, where Z_{eq} is the equivalent impedance seen by the voltage source.

For the rest of the question, assume that $L = 1 \text{ mH}$, $C_1 = 4 \text{ nF}$, $C_2 = 4 \text{ nF}$, $R_1 = 10 \text{ k}\Omega$, $R_2 = 5 \text{ k}\Omega$. **Do not** ignore C_2 in these calculations.

(c) [4 points] Find $v_{out}(t)$ if $v_{in}(t) = 3 \cos(\omega_1 t + \pi/4)$ Volts.

(d) [4 points] Find $v_{out}(t)$ if $v_{in}(t) = 3 \cos(\omega_2 t + \pi/4)$ Volts.

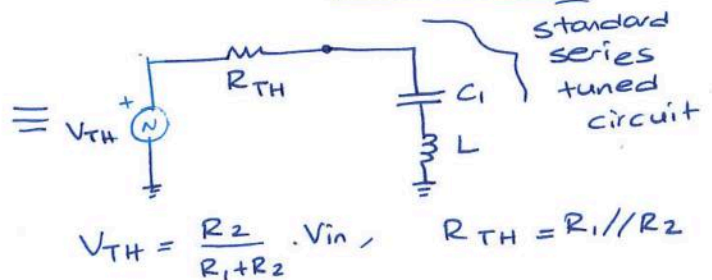
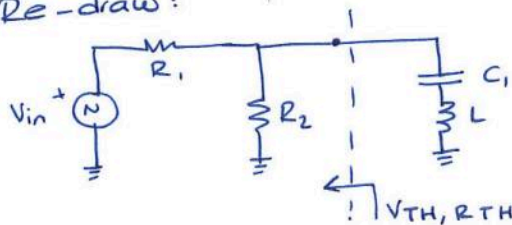
a) Ignoring C_2 :



Series tuned circuit:

$$\omega_0 = \frac{1}{\sqrt{LC_1}}$$

Re-draw:



Then,

$$Q = \frac{\omega_0 L}{R_{TH}} = \frac{\frac{1}{\sqrt{LC_1}} L}{\frac{R_1 R_2}{R_1 + R_2}} = \frac{R_1 + R_2}{R_1 R_2} \sqrt{\frac{L}{C_1}}$$

$$\begin{aligned} b) Z_{eq} &= R_1 + \left(\frac{1}{j\omega L + \frac{1}{j\omega C_1}} + j\omega C_2 + \frac{1}{R_2} \right)^{-1} = R_1 + \left(\frac{j\omega C_1}{1 - \omega^2 LC_1} + j\omega C_2 + \frac{1}{R_2} \right)^{-1} \\ &= R_1 + \left(\frac{j\omega R_2 C_1 + j\omega R_2 C_2 (1 - \omega^2 LC_1) + (1 - \omega^2 LC_1)}{R_2 (1 - \omega^2 LC_1)} \right)^{-1} \\ &= R_1 + \frac{R_2 (1 - \omega^2 LC_1)}{(1 - \omega^2 LC_1) + j\omega R_2 (C_1 + C_2 (1 - \omega^2 LC_1))} \end{aligned}$$

$\text{Im}\{Z_{eq}(\omega)\} = 0 \Rightarrow$ This is satisfied if

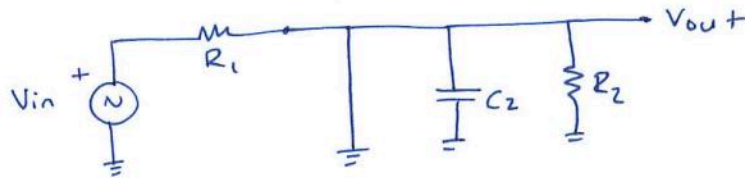
- 1) $(1 - \omega^2 LC_1) = 0$
- 2) $C_1 + C_2 (1 - \omega^2 LC_1) = 0$

$\Rightarrow$

So, $1 - \omega_1^2 LC_1 = 0 \Rightarrow \boxed{\omega_1 = \frac{1}{\sqrt{LC_1}}}$

$C_1 + C_2(1 - \omega_2^2 LC_1) = 0 \Rightarrow \omega_2^2 = \frac{C_1 + C_2}{LC_1 C_2} \Rightarrow \boxed{\omega_2 = \frac{1}{\sqrt{L \frac{C_1 C_2}{C_1 + C_2}}}}$

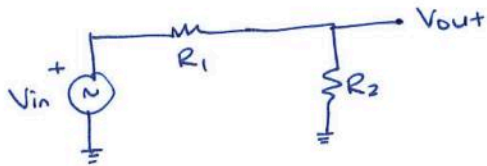
c) at ω_1 : $Z_{eq} = R_1$ (C_1 & L together act like short circuit)



So, $\boxed{V_{out}(t) = 0}$

d) at ω_2 : $Z_{eq} = R_1 + \frac{R_2(1 - \omega^2 LC_1)}{1 - \omega^2 LC_1} = R_1 + R_2$

(L, C_1, C_2 together act like open circuit)

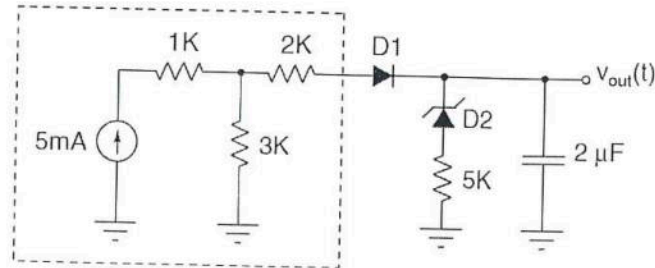


$V_{out}(t) = \frac{R_2}{R_1 + R_2} V_{in}(t)$

$V_{out}(t) = \frac{5}{15} 3 \cos(\omega_2 t + \pi/4) V$

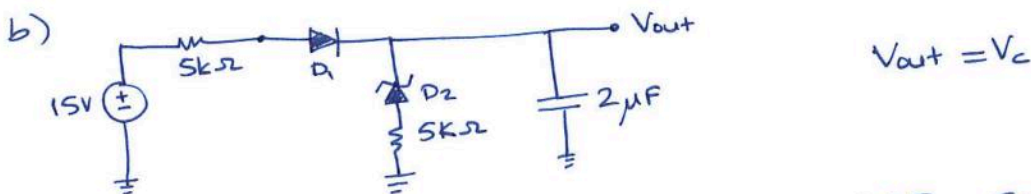
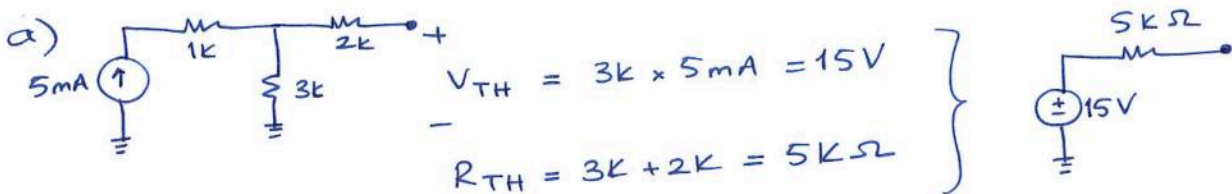
$= \boxed{\cos(\omega_2 t + \pi/4) V}$

Question 3 [25 Points]:

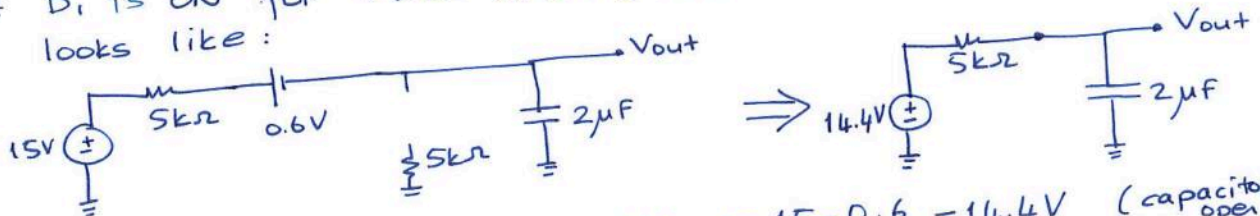


In the circuit above, the threshold voltage for both diodes is $V_{ON} = 0.6V$. The breakdown voltage for Zener diode D2 is $V_Z = 7V$. For the capacitor, $V_C(0) = 0V$.

- (a) [5 points] Find the Thevenin equivalent of the circuit inside the dashed box, as seen by the diode. Draw the Thevenin equivalent circuit.
- (b) [20 points] Find and plot $v_{out}(t)$ for $t > 0$. Clearly specify the state of each diode for all $t > 0$. Mark all important voltage levels and time points.



* D_1 is ON for $t \geq 0$. Initially, D_2 is OFF, so, the circuit looks like:



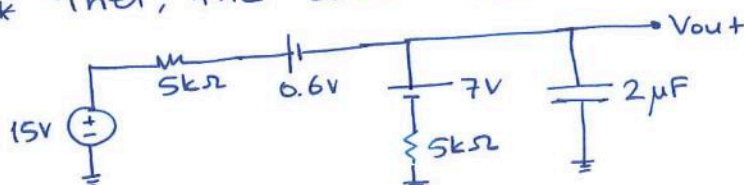
Capacitor starts charging. $V_{final} = 15 - 0.6 = 14.4V$ (capacitor open circuit)
 $R_{eq} = 5K\Omega \Rightarrow \tau_1 = R_{eq} \cdot C = 10ms$

$$V_{out}(t) = V_c(t) = V_{final} + (V_{init} - V_{final}) e^{-t/\tau} = \boxed{14.4 - 14.4 e^{-t/\tau_1} V}$$

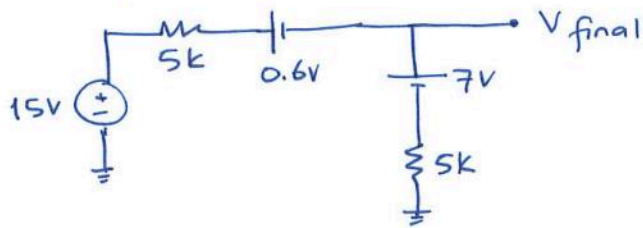
* When $V_{out}(t)$ reaches 7V, Zener diode goes into breakdown mode. This happens when:

$$V_{out}(t_1) = 7 = 14.4 - 14.4 e^{-t_1/\tau_1} \Rightarrow \boxed{t_1 = 6.66ms}$$

* Then, the circuit looks like:



* To find V_{final} : Capacitor $\Rightarrow$ open circuit

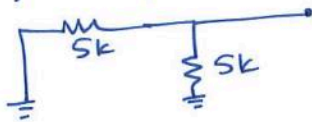


KCL @ V_{final}

$$\frac{V_{final} - (15 - 0.6)}{5k} + \frac{V_{final} - 7}{5k} = 0$$

$$V_{final} = 10.7V$$

* To find R_{eq} : Kill all sources



$$R_{eq} = \frac{5}{2} k\Omega, \text{ so } \tau_2 = \frac{5}{2} k\Omega \times 2\mu f = 5ms$$

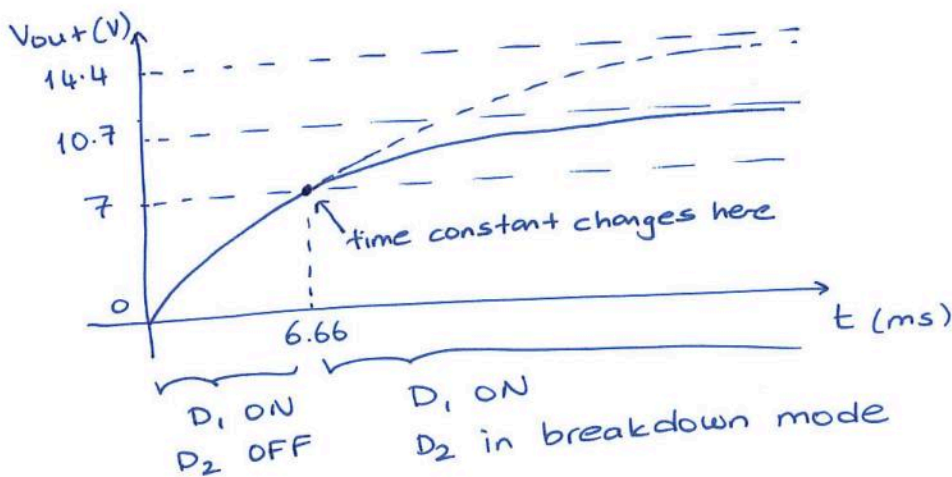
* Then,

$$\begin{aligned} V_{out}(t) = V_c(t) &= V_{final} + (V_{init} - V_{final}) e^{-(t-t_1)/\tau_2} \\ &= 10.7 + (7 - 10.7) e^{-(t-t_1)/\tau_2} \text{ V} \\ &= 10.7 - 3.7 e^{-(t-t_1)/\tau_2} \text{ V} \end{aligned}$$

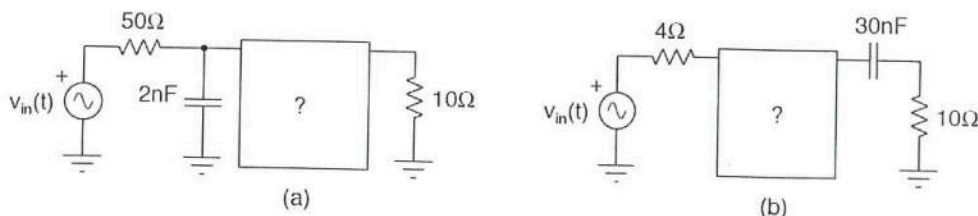
* So, overall:

$$V_{out}(t) = \begin{cases} 14.4 - 14.4 e^{-t/10} \text{ V}, & 0 < t < 6.66 \text{ ms} \\ 10.7 - 3.7 e^{-(t-6.66)/5} \text{ V}, & t > 6.66 \text{ ms} \end{cases}$$

where t is in ms



Question 4 [25 Points]:



In the circuits above, $v_{in}(t) = 2 \cos(10^7 t + \pi/4)$ Volts. For both (a) and (b), we would like to design the circuit block marked as "?" to maximize the power transfer to the 10Ω load.

In your solutions, clearly specify the component values for all elements, and also specify the number of turns for inductors/transformers (you can leave the number of turns as non-integer numbers). Assume that we have a core with $A_L = 40 \text{ nH/turns}^2$.

Simplify your design to reduce the number of components as much as possible.

(a) [12 points] Design the block marked as "?" for the circuit in (a). Draw the overall circuit. Compute the power dissipated in the 10Ω load in the case of maximum power transfer.

(b) [13 points] Design the block marked as "?" for the circuit in (b). Draw the overall circuit. Compute the power dissipated in the 10Ω load in the case of maximum power transfer.

a) many potential solutions.

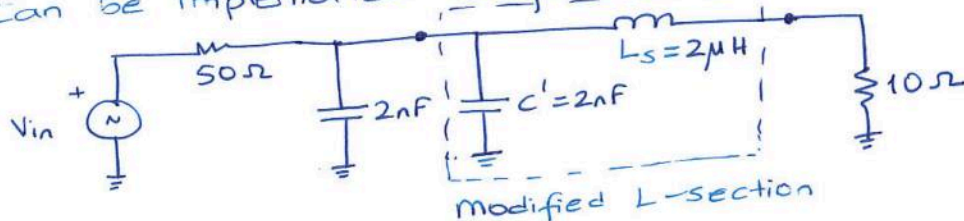
Solution 1: L-section. First, ignore 2nF capacitor.

$$R_p = (Q^2 + 1)R_s \Rightarrow Q = \sqrt{\frac{50}{10} - 1} = 2$$

$$Q = \frac{\omega_0 L_s}{R_s} \Rightarrow L_s = \frac{2 \cdot 10}{10^7} = 2 \mu\text{H} \Rightarrow n = \sqrt{\frac{L_s}{A_L}} = 7.07 \text{ turns}$$

$$\omega_0 = \frac{1}{\sqrt{L_s C (1 + \frac{1}{Q^2})}} \Rightarrow C = \frac{1}{\omega_0^2 L_s (1 + \frac{1}{Q^2})} = \frac{1}{10^{14} \cdot 2 \cdot 10^{-6} \cdot (1 + \frac{1}{4})} = 4 \text{ nF}$$

Can be implemented as follows:



$$C = C' + 2\text{nF} = 4\text{nF} \Rightarrow C' = 2\text{nF}$$

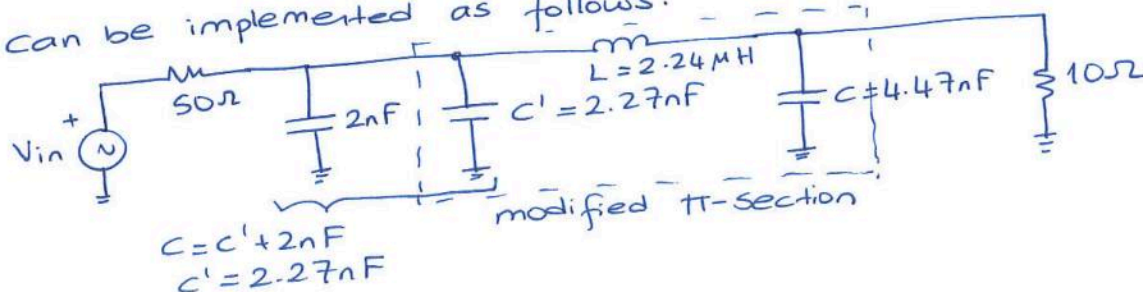
Solution 2: Π -section impedance inverter. First, ignore 2nF capacitor.

$$Z_I = \frac{X^2}{Z} \Rightarrow X^2 = \sqrt{50 \cdot 10} = 10\sqrt{5}$$

$$jX = j\omega_0 L \Rightarrow L = \frac{X}{\omega_0} = \frac{10\sqrt{5}}{10^7} = 2.24 \mu\text{H} \Rightarrow n = \sqrt{\frac{L}{A_L}} = 7.48 \text{ turns}$$

$$-jX = \frac{1}{j\omega_0 C} \Rightarrow C = \frac{1}{X \omega_0} = \frac{1}{10\sqrt{5} \cdot 10^7} = 4.47 \text{ nF}$$

Can be implemented as follows:

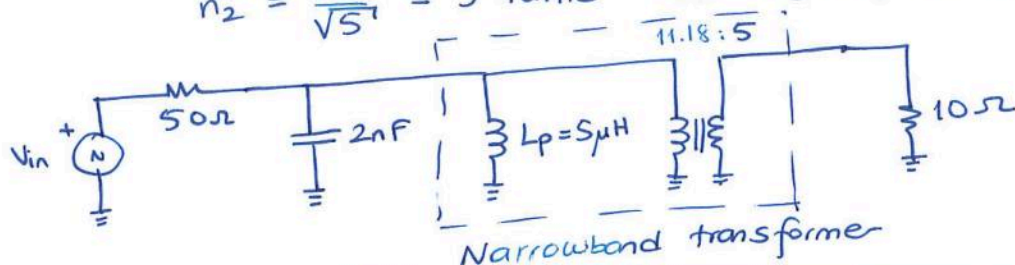


Solution 3: Transformer. Choose L_p to tune out 2 nF capacitor.

$$L_p = \frac{1}{\omega_0^2 C} = \frac{1}{10^{14} \cdot 2 \cdot 10^{-9}} = 5 \mu\text{H} \Rightarrow n_1 = \sqrt{\frac{L_p}{A_L}} = 11.18 \text{ turns}$$

Then, $R_s = \left(\frac{n_1}{n_2}\right)^2 R_L \Rightarrow \frac{n_1}{n_2} = \sqrt{\frac{50}{10}} = \sqrt{5}$

$$n_2 = \frac{n_1}{\sqrt{5}} = 5 \text{ turns} \Rightarrow L_s = n_2^2 A_L = 1 \mu\text{H}$$



For maximum power transfer: $P_{\max} = P_A = \frac{V_s^2}{8R_s} = \frac{2^2}{8 \cdot 50} = 0.01 \text{ Watt} = \boxed{10 \text{ mW}}$

b) Many potential solutions.

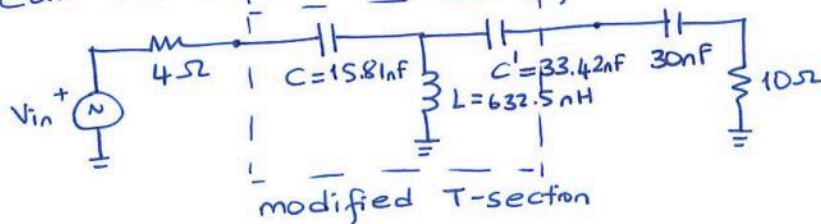
Solution 1: T-section impedance inverter. First, ignore 30 nF capacitor.

$$4 = \frac{X^2}{10} \Rightarrow X = \sqrt{40} = 2\sqrt{10}$$

$$jX = j\omega_0 L \Rightarrow L = \frac{X}{\omega_0} = \frac{2\sqrt{10}}{10^7} = 632.5 \text{ nH} \Rightarrow n = \sqrt{\frac{L}{A_L}} = 3.98 \text{ turns}$$

$$-jX = \frac{1}{j\omega_0 C} \Rightarrow C = \frac{1}{X\omega_0} = \frac{1}{2\sqrt{10} \cdot 10^7} = 15.81 \text{ nF}$$

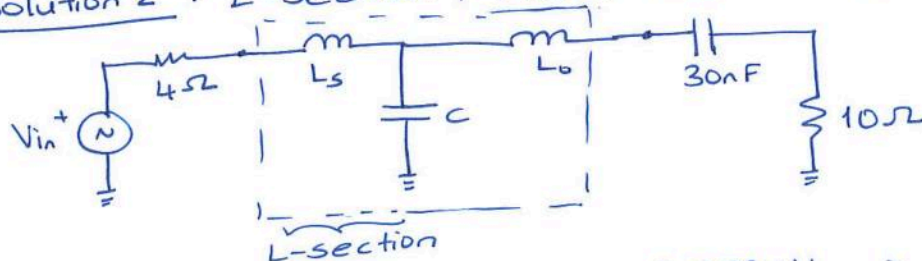
Can be implemented as follows:



$$\frac{C' \cdot 30}{C' + 30} = 15.81$$

$$C' = 33.42 \text{ nF}$$

Solution 2: L-section. First, tune out 30 nF capacitor with L_o



$$L_o = \frac{1}{\omega_0^2 \cdot 30\text{ nF}} = \frac{1}{10^{14} \cdot 30 \times 10^{-9}} = 0.333 \mu\text{H} \Rightarrow n = \sqrt{\frac{L}{A_L}} = 2.89 \text{ turns}$$

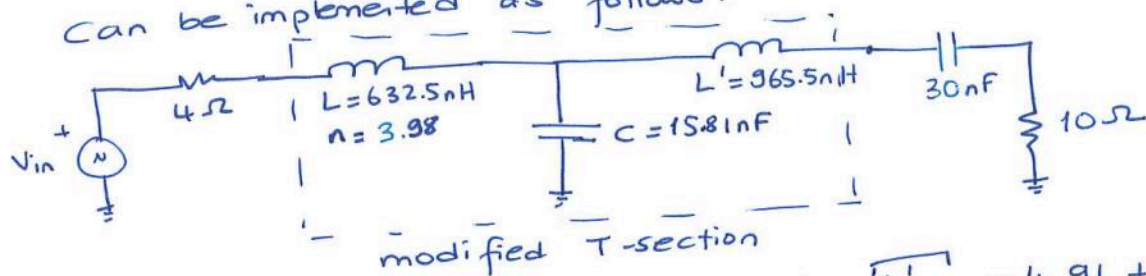
Then, $10 = (Q^2 + 1) \cdot 4 \Rightarrow Q = \sqrt{\frac{3}{2}}$

$$Q = \frac{\omega_0 L_s}{R_s} \Rightarrow L_s = \frac{R_s \cdot Q}{\omega_0} = \frac{4 \cdot \sqrt{\frac{3}{2}}}{10^7} = 0.49 \mu\text{H} \Rightarrow n_s = \sqrt{\frac{L_s}{A_L}} = 3.5 \text{ turns}$$

$$\omega_0 = \frac{1}{\sqrt{L_s C (1 + \frac{1}{Q^2})}} \Rightarrow C = \frac{1}{\omega_0^2 L_s (1 + \frac{1}{Q^2})} = 12.24 \text{ nF}$$

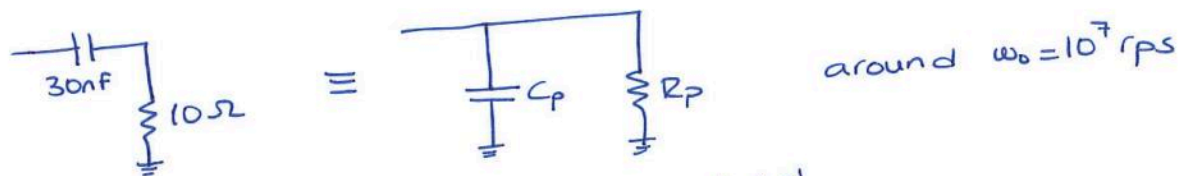
Solution 3: T-section. First tune out 30 nF capacitor with
 $L = 632.5 \text{ nH}$
 $C = 15.81 \text{ nF}$ } see previous page, (Solution 1.)
 $L_0 = 0.333 \mu\text{H}$ (see previous page)

Can be implemented as follows:



modified T-section
 $L' = L + L_0 = 965.5 \text{ nH} \Rightarrow n' = \sqrt{\frac{L'}{A_L}} = 4.91 \text{ turns}$

Solution 4: Transformer. First, find parallel equivalent of 30 nF in series with 10 ohm



$$Z_{eq} = \frac{1}{j10^7 \cdot 30 \cdot 10^{-9}} + 10 = \left(j\omega_0 C_p + \frac{1}{R_p} \right)^{-1}$$

$$-j\frac{10}{3} + 10 = \frac{R_p}{1 + j\omega_0 R_p C_p} \frac{(1 - j\omega_0 R_p C_p)}{(1 - j\omega_0 R_p C_p)}$$

$$10 - j\frac{10}{3} = \frac{R_p}{1 + \omega_0^2 R_p^2 C_p^2} - \frac{j\omega_0 R_p^2 C_p}{1 + \omega_0^2 R_p^2 C_p^2}$$

Real parts and imaginary parts must match

$$* \quad 10 = \frac{R_p}{1 + \omega_0^2 R_p^2 C_p^2} \Rightarrow 1 + \omega_0^2 R_p^2 C_p^2 = \frac{R_p}{10}$$

$$\frac{10}{3} = \frac{\omega_0 R_p^2 C_p}{1 + \omega_0^2 R_p^2 C_p^2} \Rightarrow \frac{10}{3} = \frac{\omega_0 R_p^2 C_p}{R_p/10} \Rightarrow \omega_0 R_p C_p = \frac{1}{3}$$

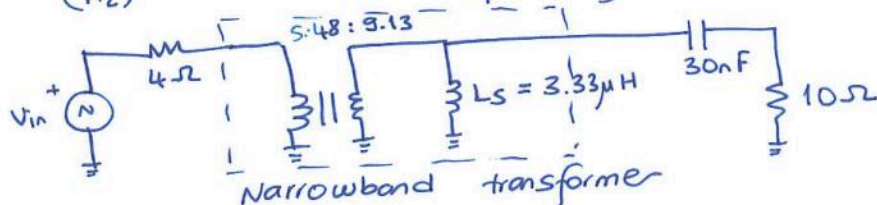
$$\text{Then, } 1 + \left(\frac{1}{3}\right)^2 = \frac{R_p}{10} \Rightarrow R_p = \frac{100}{9} = 11.11 \Omega$$

$$C_p = \frac{1}{\omega_0 R_p} \cdot \frac{1}{3} = \frac{1}{10^7 \cdot 100/9} \cdot \frac{1}{3} = 3 \text{ nF}$$

* Then, choose L_s to tune out C_p :

$$L_s = \frac{1}{\omega_0^2 C_p} = \frac{1}{10^{14} \cdot 3 \cdot 10^{-9}} = 3.33 \mu\text{H} \Rightarrow n_2 = \sqrt{\frac{L_s}{A_L}} = 9.13 \text{ turns}$$

$$* \quad 4 = \left(\frac{n_1}{n_2}\right)^2 \cdot R_p \Rightarrow \frac{n_1}{n_2} = \sqrt{\frac{4}{R_p}} = \frac{3}{5} \Rightarrow n_1 = \frac{3}{5} \cdot n_2 = 5.48 \text{ turns}$$



For maximum power transfer: $P_{\max} = P_A = \frac{V_s^2}{8R_s} = \frac{4}{8.4} = \underline{\underline{0.125 \text{ W}}}$